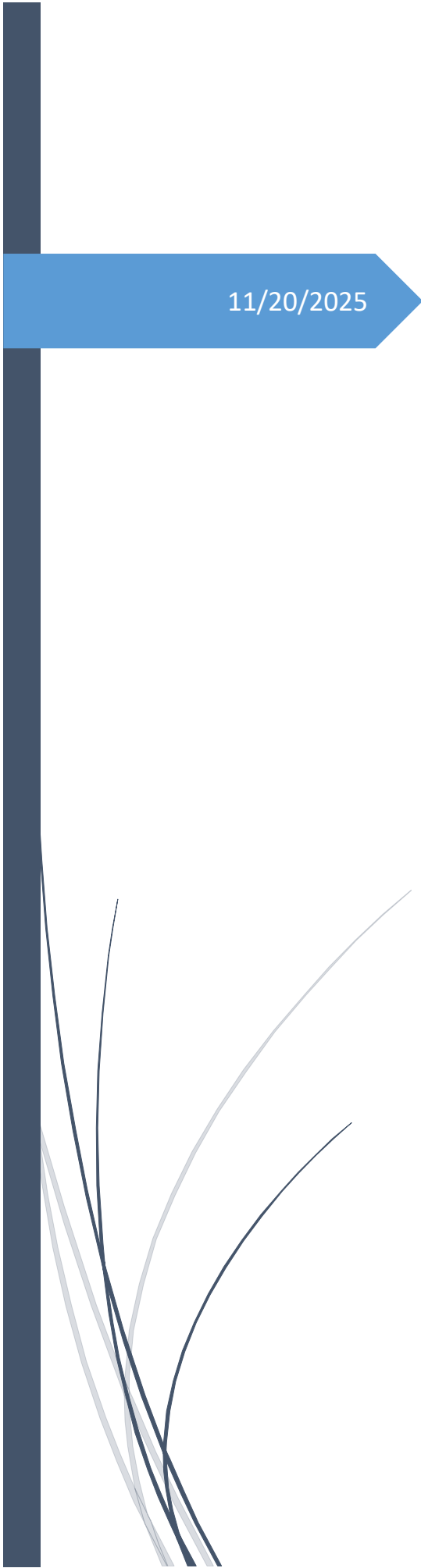




11/20/2025

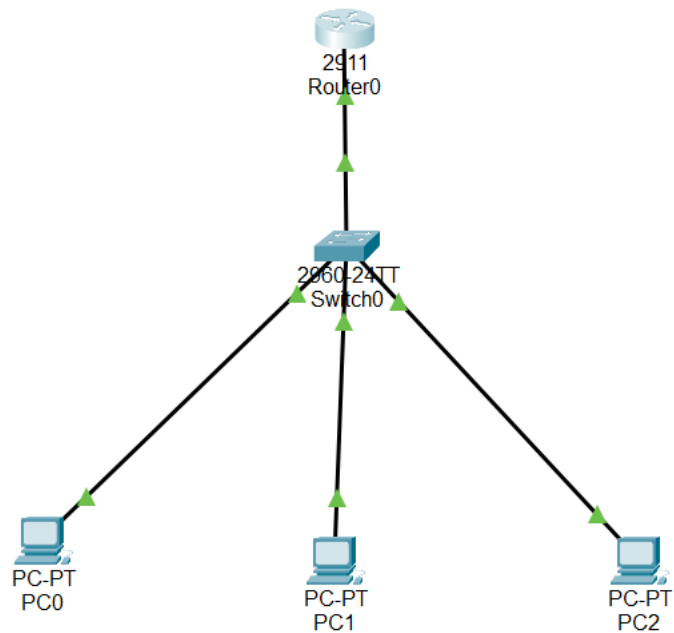
COMPUTER NETWORK

LAB 14

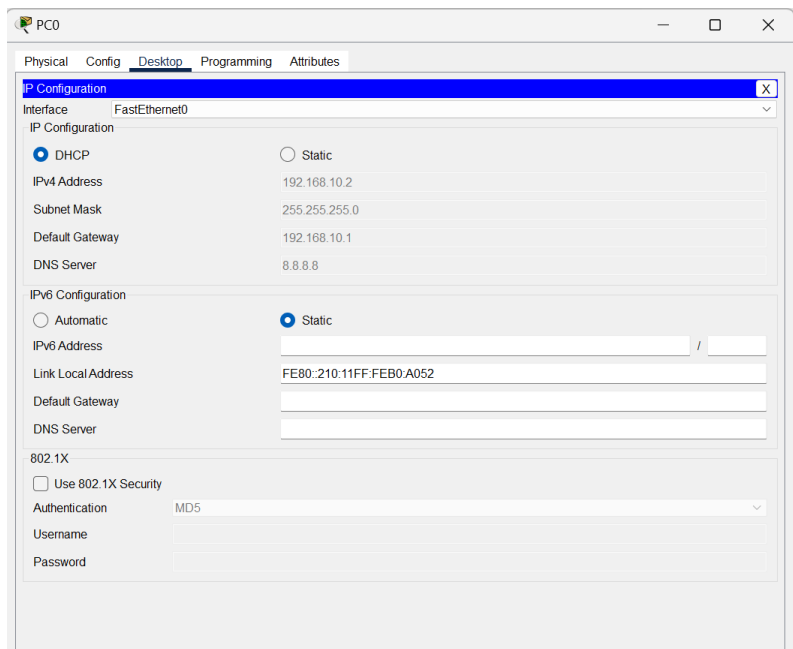
IFZA RIZWAN
SAP 64281

TASK 1:

Topology created:



PC 0, PC 1, PC 2:



COMPUTER NETWORK

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.10.6

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20A:F3FF:FE2C:96B0

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.10.7

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FE43:DD8

Default Gateway

DNS Server

802.1X

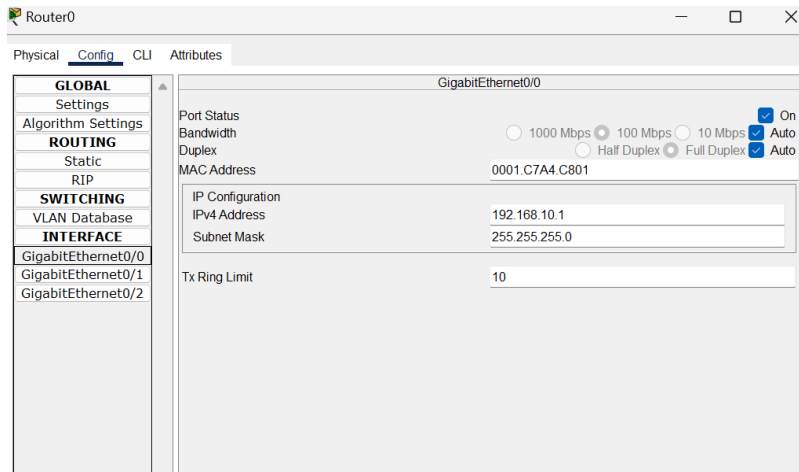
☐ Use 802.1X Security

Authentication MD5

Username

Password

ROUTER CONFIGURATION:



CLI tab:

```

Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)# ip dhcp excluded-address 192.168.10.2 192.168.10.5
^
% Invalid input detected at '^' marker.

Router(config-if)#Router(config)# ip dhcp pool LAN-POOL
^
% Invalid input detected at '^' marker.

Router(config-if)#Router(dhcp-config)# network 192.168.10.0 255.255.255.0
^
% Invalid input detected at '^' marker.

Router(config-if)#Router(dhcp-config)# default-router 192.168.10.1
^
% Invalid input detected at '^' marker.

Router(config-if)#Router(dhcp-config)# dns-server 8.8.8.8
^
% Invalid input detected at '^' marker.

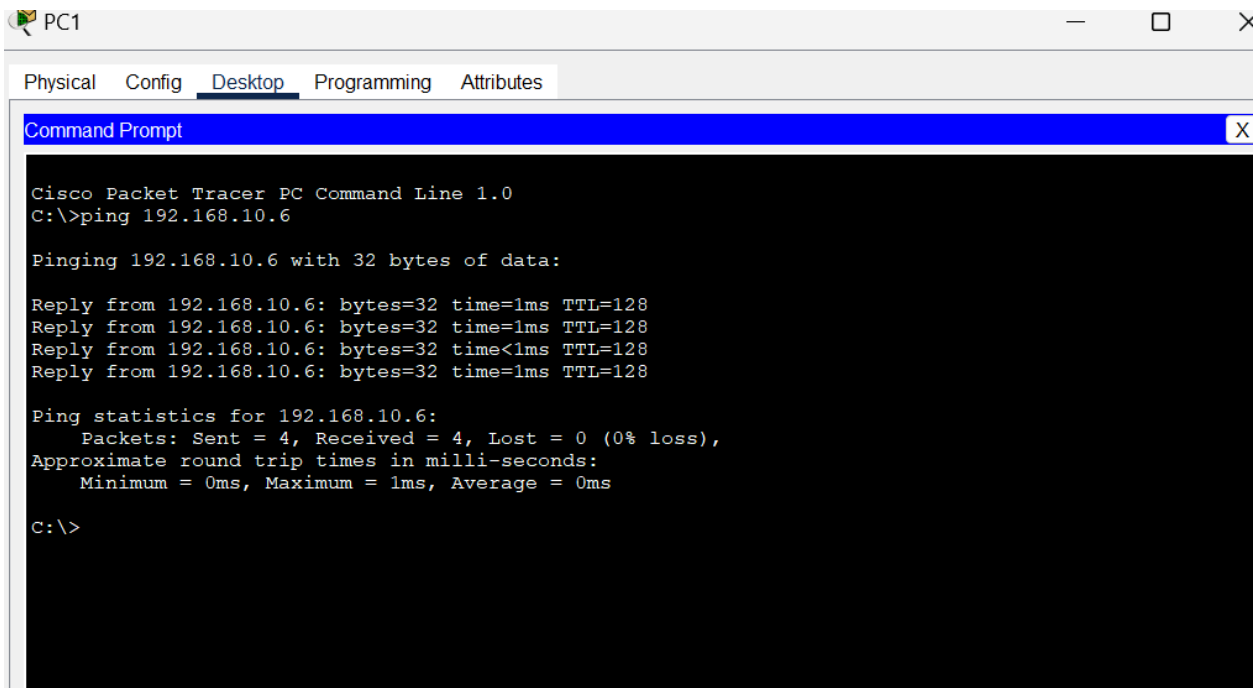
Router(config-if)#Router(dhcp-config)# exit

```

```

^
% Invalid input detected at '^' marker.
Router(config-if)#Router(config)# interface gigabitEthernet 0/0
^
% Invalid input detected at '^' marker.
Router(config-if)#Router(config-if)# ip address 192.168.10.1 255.255.255.0
^
% Invalid input detected at '^' marker.
Router(config-if)#Router(config-if)# no shutdown
^
% Invalid input detected at '^' marker.
Router(config-if)#Router(config-if)# exit
^
% Invalid input detected at '^' marker.
Router(config-if)#Router# write memory
^
% Invalid input detected at '^' marker.
Router(config-if)#ip dhcp excluded-address 192.168.10.2 192.168.10.5
Router(config)#
Router(config)#ip dhcp pool LAN-POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#
Router(dhcp-config)#interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#

```



PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

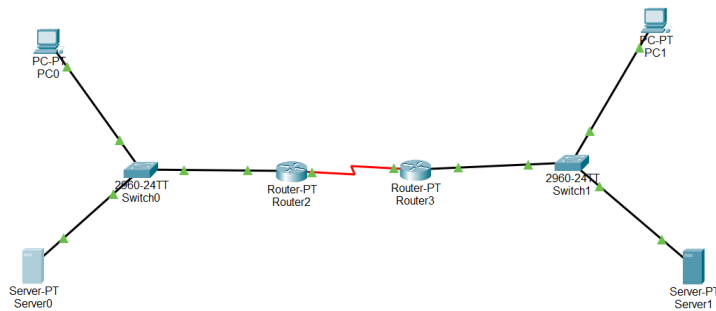
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

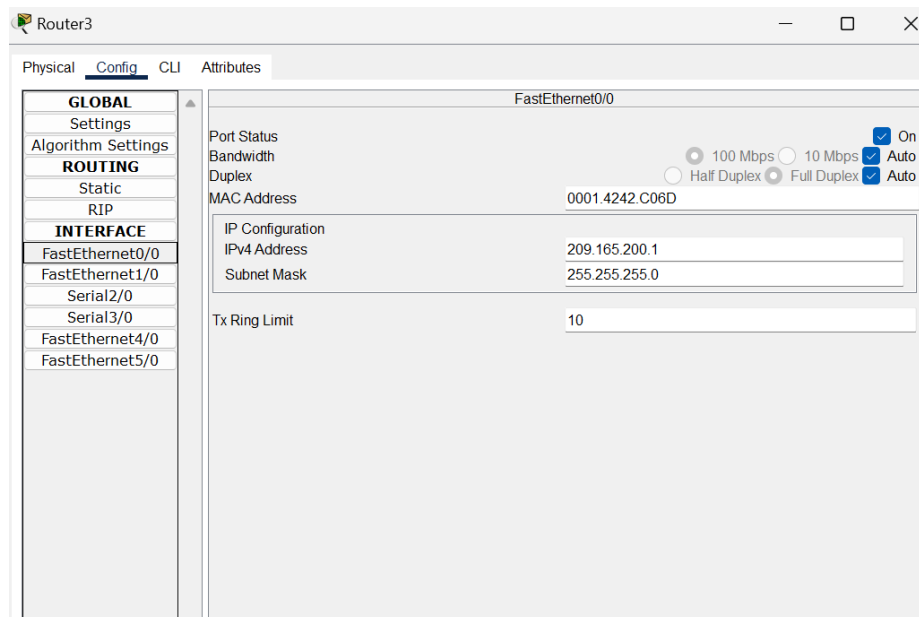
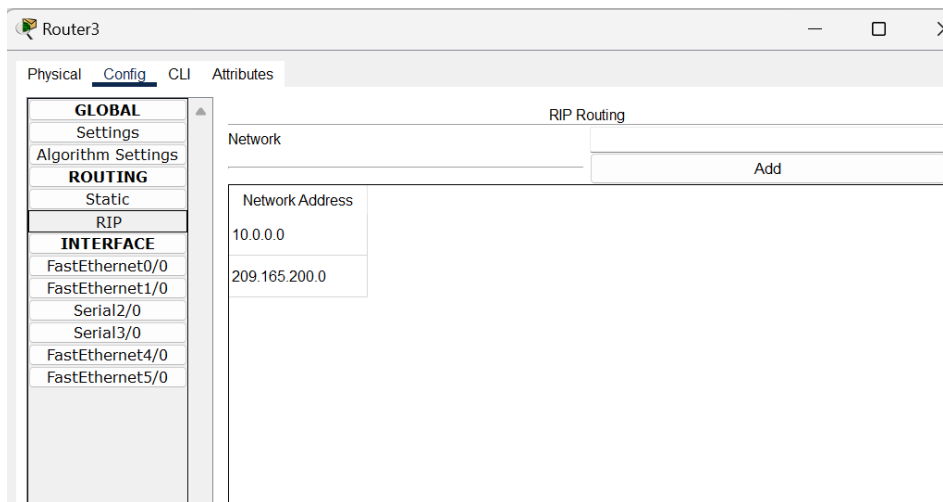
C:\>

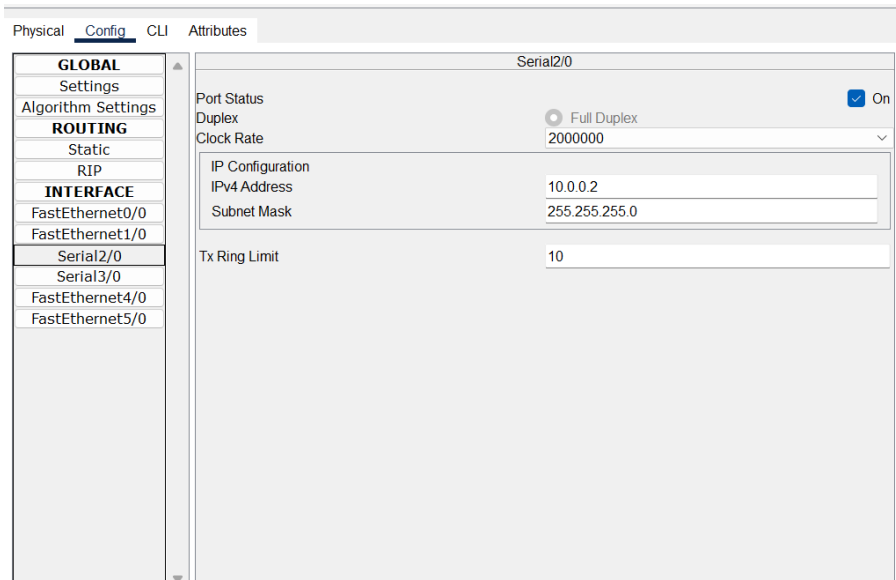
```

TASK 2:



Configuration of router 3:





CLI Configuration: this is configuration of default route and dhcp enable other configuration is in the above pictures.

```

Router(config)#!
Router(config)#ip dhcp pool ISP_POOL
Router(dhcp-config)# network 209.165.200.0 255.255.255.0
Router(dhcp-config)# default-router 209.165.200.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)#!
Router(dhcp-config)#ip dhcp excluded-address209.165.200.1
      ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console
no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#Router2(config)# ip route 0.0.0.0 0.0.0.0 <gateway_ip_R2>
      ^
% Invalid input detected at '^' marker.

Router(config-if)#Router2(config)# ipv6 route ::/0 <gateway_ipv6_R2>
      ^
% Invalid input detected at '^' marker.

Router(config-if)#write memory
      ^
% Invalid input detected at '^' marker

```

Configuration of router 2:

Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

RIP Routing

Network

Add

Network Address

10.0.0.0

192.168.1.0

Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status

Bandwidth

Duplex

MAC Address

IP Configuration

IPv4 Address

Subnet Mask

Tx Ring Limit

100 Mbps

10 Mbps

Half Duplex

Full Duplex

On

Auto

Auto

0001.9676.4C17

192.168.1.1

255.255.255.0

10

Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status

Duplex

Clock Rate

IP Configuration

IPv4 Address

Subnet Mask

Tx Ring Limit

Full Duplex

1200

On

10.0.0.1

255.255.255.0

10

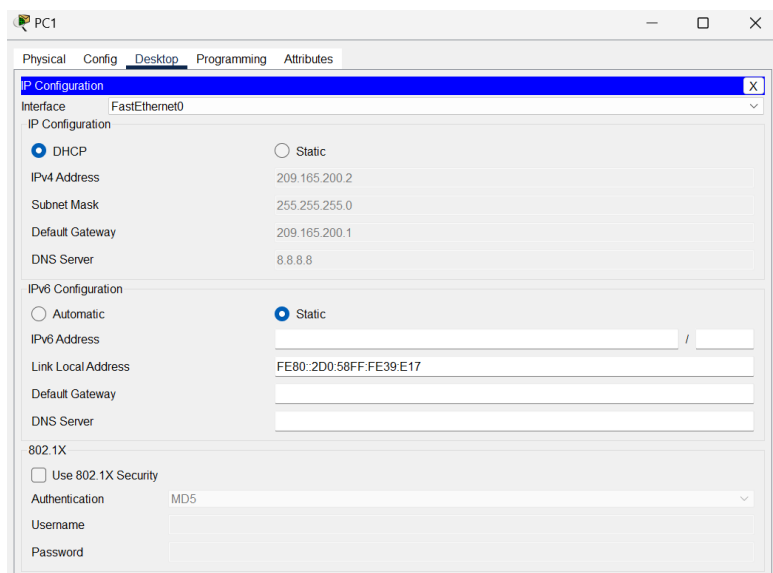
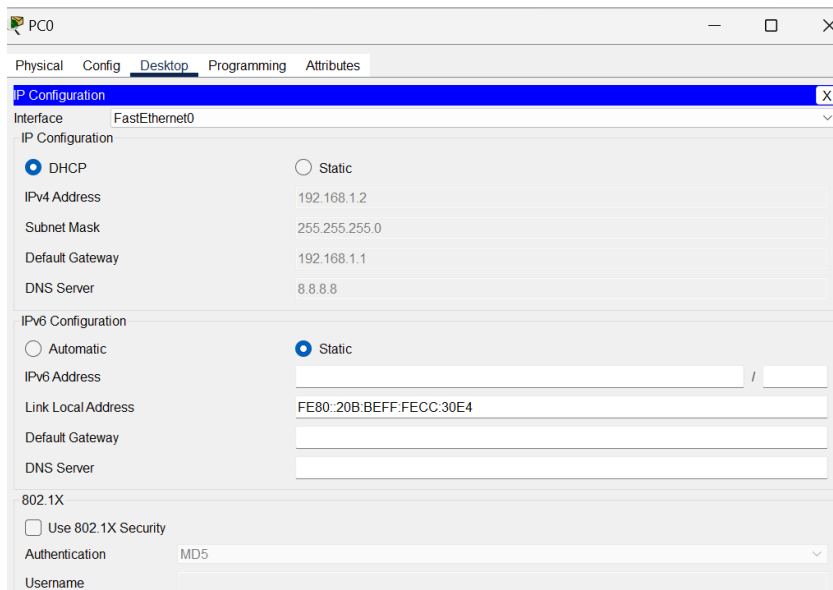
Default Route config in CLI:

```

INEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config-if)#Router1(config)# ip route 0.0.0.0 0.0.0.0 <gateway_ip_R1>
      ^
Invalid input detected at '^' marker.
Router(config-if)#Router1(config)# ipv6 route ::/0 <gateway_ipv6_R1>
      ^

```

Now enable DHCP in each server and pc :



COMPUTER NETWORK

Server1

Physical Config Services Desktop Programming Attributes

IP Configuration [X]

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 209.165.200.3

Subnet Mask 255.255.255.0

Default Gateway 209.165.200.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80:20B:BEFF:FE63:1A89

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration [X]

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80:205:5EFF:FE45:8714

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Ping from server 0 to pc 1 and server 1:

```
C:\>ping 209.165.200.3

Pinging 209.165.200.3 with 32 bytes of data:

Reply from 209.165.200.3: bytes=32 time=1ms TTL=126
Reply from 209.165.200.3: bytes=32 time=11ms TTL=126
Reply from 209.165.200.3: bytes=32 time=10ms TTL=126
Reply from 209.165.200.3: bytes=32 time=11ms TTL=126

Ping statistics for 209.165.200.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 8ms

C:\>ping 209.165.200.2

Pinging 209.165.200.2 with 32 bytes of data:

Request timed out.
Reply from 209.165.200.2: bytes=32 time=2ms TTL=126
Reply from 209.165.200.2: bytes=32 time=10ms TTL=126
Reply from 209.165.200.2: bytes=32 time=10ms TTL=126

Ping statistics for 209.165.200.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 7ms

C:\>
```

THE END